

# Music Emotion Classification Using Spotify Audio Features for Mood-Based Recommendation Systems

Maharin Sharif Abony - 22299442

Afra Nawar - 23301633

Ramisa Fariha Islam - 24101664

Department of Computer Science and Engineering, BRAC University

**Abstract**—Digital music streaming platforms mainly rely on metadata such as genre, artist, album, and popularity for recommendation, which often fails to capture the emotional intention of users. Since listeners frequently search for music based on mood, emotion-aware recommendation systems have become an important research area in Music Information Retrieval (MIR). This paper presents a machine learning based approach for classifying Spotify songs into four emotional categories: Happy, Sad, Energetic, and Calm. The study uses the Moodify Dataset containing approximately 278,000 Spotify songs labeled with emotions and multiple Spotify-generated audio features such as danceability, energy, loudness, valence, tempo, acousticness, liveness, and instrumentalness. Exploratory Data Analysis (EDA) was conducted to understand class distributions and feature relationships. Preprocessing included duplicate removal, irrelevant feature elimination, class balancing using undersampling, outlier handling using IQR-based capping, and feature scaling using StandardScaler. Multiple machine learning and deep learning models including Logistic Regression, Random Forest, LightGBM, Dense Neural Network (DNN), XGBoost, and TabNet were trained and evaluated using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC. Experimental results show that XGBoost achieved the best overall performance with the highest classification accuracy and robustness. The findings confirm that Spotify audio features are highly effective for emotion recognition and can support real-world mood-based music recommendation systems.

**Index Terms**—Music Emotion Recognition, Machine Learning, Spotify Audio Features, Mood Classification, XGBoost, Recommendation System

## I. INTRODUCTION

The emergence of the online music streaming industry, including Spotify, has revolutionized music discovery but also has continued to pose a challenge in finding tracks that resonate with the emotional state of a listener. Traditional recommendation systems are based on metadata genre, artist, and listening history, which do not reflect the affective aspect of music. This is a big gap in that music is well used to control emotions, relax, provide therapeutic intervention, and entertainment. One of the most important remarks is that emotional perception does not follow any genre: the songs of different genres can produce the same mood, and songs of the same genre can trigger opposite emotional reactions. This is the focus of a subfield of Music Information Retrieval (MIR), the Music Emotion Recognition (MER). The quantitative audio characteristics (valence, energy, danceability, loudness, acousticness, speechiness, and tempo) of Spotify can be regarded as a rich numerical representation that is highly appropriate in terms of supervised emotion classification.

In this study, a supervised machine learning model is developed to classify songs into four moods Happy, Sad, Energetic, and Calm using Spotify audio features. They are trained and compared to six classifiers: Logistic Regression, Random Forest, LightGBM, XGBoost, a Dense Neural Network (DNN), and TabNet. The study performs exploratory data analysis to determine feature-emotion relationships, preprocesses data, including outlier treatment, StandardScaler normalization and random undersampling to balance the classes, and evaluates the models based on accuracy, precision, recall, F1-score, confusion matrix, and ROC/AUC - with the aim of determining the best-performing classifier in mood-based music recommendation.

### A. Problem Statement

The primary problem with this initiative is the subjectivity of music discovery. Despite the vast variety of music available in digital libraries (millions of songs), standard metadata (genre or artist) does not reflect a certain emotional state that a listener may be attempting to achieve. The perceived emotional impact of audio differs significantly from technical information about it, such as its pace or acousticness. By using the raw audio features of the Spotify songs, the proposed project will assist close this gap and develop a machine learning model that can accurately predict the four human moods of Happy, Sad, Energetic, and Calm.

## II. RELATED WORK

Early Music Emotion Recognition (MER) research mainly focused on traditional machine learning using handcrafted audio features such as valence, energy, tempo, danceability, loudness, acousticness, and speechiness. Panda et al. explored classifiers including SVM, KNN, Naive Bayes, Decision Trees, and Random Forest, showing that rhythm, melody, harmony, timbre, and expressivity strongly influence emotion recognition [1]. Dalida et al. used Logistic Regression with Spotify audio features to classify emotions into Happy, Sad, Calm, and Energetic, achieving about 96% accuracy, while identifying energy, valence, acousticness, and speechiness as key predictors [4]. Similarly, Álvarez et al. found Random Forest to be the best-performing classifier in the RIADA framework [10].

Ensemble learning methods further improved performance. Supraba and Sunyoto applied Random Forest with SMOTE balancing on 171,819 Spotify songs and achieved 98.77% accuracy [2]. Visutsak et al. also demonstrated strong results

using Random Forest and XGBoost in mood-based recommendation systems [5].

Recent studies shifted toward deep learning and multimodal approaches. Talaghat et al. combined audio and lyrics using CNN, LSTM, BERT, and XLNet, showing multimodal fusion outperformed unimodal systems [3]. Hao et al. proposed a Bi-LSTM multimodal framework integrating audio, video, and physiological signals for improved real-time emotion recognition [6].

Large-scale datasets have also become important for MER advancement. DISCO-10M [7], PDMX [9], and multimodal datasets discussed by Christodoulou et al. [8] improved model generalization and reproducibility. Patil et al. summarized current MER challenges, including emotion subjectivity and the lack of standardized datasets [11].

In this study, we use the large-scale Moodify Dataset (~278,000 labeled Spotify songs) and compare multiple traditional and deep learning models, including Logistic Regression, Random Forest, LightGBM, XGBoost, Dense Neural Network, and TabNet, to identify the most effective approach for music emotion classification.

### III. DATASET DESCRIPTION

#### A. Dataset Preparation

This study used the Moodify Dataset - 278k Emotion Labeled Spotify Songs dataset that is available on Kaggle. This dataset includes around 278,000 songs from Spotify, each labeled into four emotions: Happy, Sad, Energetic, and Calm. It contains several numerical audio features generated by Spotify which describe acoustic and musical properties. From the EDA part (correlation matrix, label distribution, feature distribution, box-plot, histogram, KDE, pairplot etc.) the decision of data pre-processing was taken.

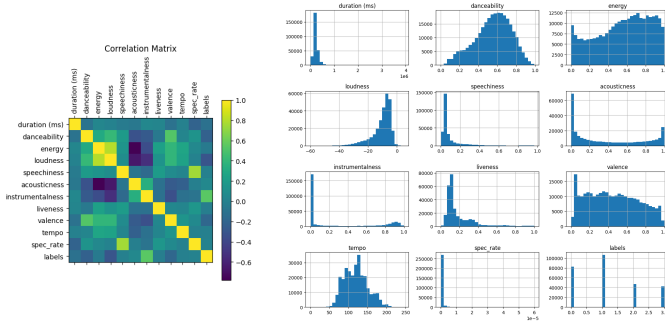


Fig. 1. Exploratory Data Analysis including correlation matrix and feature distributions

#### B. Data Preprocessing

1) *Removal of Irrelevant Features and Duplicates:* The columns Unnamed: 0, Unnamed: 0.1, and uri were removed because they do not have a direct impact on predictions and may introduce model noise. Duplicate rows were also removed to avoid learning from repeated data points.

2) *Class Balancing:* The dataset was unbalanced in terms of the four classes of emotions. Random undersampling was used to equalise the number of samples in each class. This enhanced the fairness of classification performance.

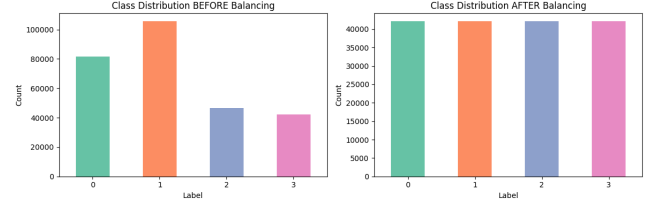


Fig. 2. Class distribution before and after balancing

3) *Outlier Handling:* The Interquartile Range (IQR) method was applied to handle outliers. Data points above the upper bound and below the lower bound were truncated instead of deleted to maintain the number of data points and minimize the impact of extreme values.

4) *Feature Scaling:* Numerical features were normalized using StandardScaler to ensure that all features had equal influence during training.

5) *Train-Test Split:* The refined dataset was split into training and testing data using an 80:20 ratio, where 80% was used for training and 20% was used for testing.

### IV. METHODOLOGY

This study employs a supervised machine learning framework for multiclass emotion classification of Spotify tracks using the Moodify dataset, where the target variable encodes four emotional categories — Happy, Sad, Energetic, and Calm — and input features were selected based on prior literature and their empirical correlation with music-induced affect. Prior to model training, a series of preprocessing steps were applied, including removal of non-predictive columns, duplicate elimination, random undersampling for class balance, IQR-based outlier truncation, StandardScaler normalization, and an 80/20 train-test split. Six classifiers spanning traditional machine learning and deep learning paradigms were then trained and compared: Logistic Regression baseline, Random Forest, LightGBM, XGBoost, a Dense Neural Network DNN, and TabNet. Model performance was evaluated using Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC/AUC to assess both aggregate and class-wise predictive performance.

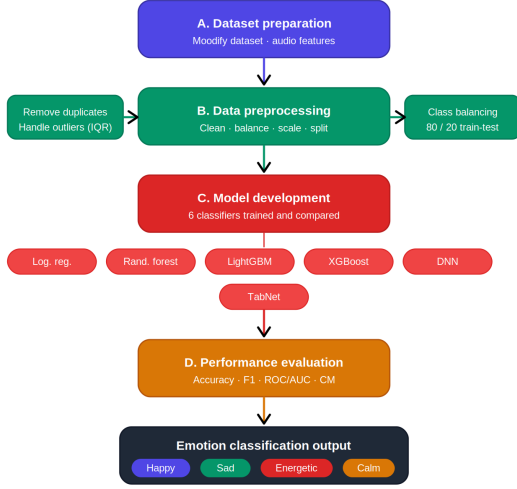


Fig. 3. Overall methodology framework for mood-based music emotion classification

## V. RESULT AND EVALUATION

To assess the performance of the deployed machine learning models for music emotion classification, several metrics were employed such as Accuracy, Precision, Recall, F1-score, Confusion Matrix, ROC-AUC Curve, Feature Importance, and Overfitting Check. The models used for comparison in this project were Logistic Regression, Random Forest, LightGBM, Dense Neural Network, XGBoost, and TabNet.

The data was initially balanced through undersampling to ensure balanced representation of all classes, then outliers were handled using IQR-based capping and features were scaled with StandardScaler. These steps helped to stabilise and improve the fairness of the models.

### A. Evaluation Metrics Framework

Multiple complementary metrics were used to comprehensively evaluate model performance.

1) *Accuracy*: The proportion of all correct predictions (both positive and negative) out of the total predictions made.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

XGBoost achieved the highest accuracy, meaning it produced the greatest number of correct predictions on the test set.

2) *Precision*: The proportion of predicted positive cases that are actually positive. It measures how many of the model's positive predictions were correct while penalising false positives.

$$\text{Precision} = \frac{TP}{TP + FP}$$

XGBoost had the best precision and was very effective at avoiding false positives.

3) *Recall (Sensitivity)*: The proportion of actual positive cases that were correctly identified by the model. Recall is also known as Sensitivity or True Positive Rate and penalises false negatives.

$$\text{Recall} = \frac{TP}{TP + FN}$$

XGBoost achieved the highest recall and identified most of the true positive cases with fewer false negatives.

4) *F1-Score*: The harmonic mean of Precision and Recall. It balances both metrics and is especially useful when classes are imbalanced.

$$\begin{aligned} \text{F1-Score} &= \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \\ &= \frac{2TP}{2TP + FP + FN} \end{aligned}$$

XGBoost again achieved the best F1-Score, demonstrating consistently high precision and recall, making it the most stable model.

### 5) ROC-AUC:

- 0.50 = Random
- 0.70 = Fair
- 0.80 = Good
- 0.90+ = Excellent

ROC-AUC is threshold-independent and performs well with imbalanced data.

**XGBoost Result: Excellent (0.95+ across all classes)**

### B. Accuracy Comparison

The accuracy of XGBoost was the highest, followed by LightGBM and Random Forest. This indicates that XGBoost was able to capture complex relationships in the audio features. Logistic Regression had the lowest accuracy as it is not the best model for non-linear classification.

TABLE I  
MODEL PERFORMANCE RANKING.

| Rank | Model                | Accuracy       | F1-Score  | Status      |
|------|----------------------|----------------|-----------|-------------|
| 1    | XGBoost              | Highest-0.94   | Highest   | BEST        |
| 2    | LightGBM             | Very High-0.91 | Very High | Alt. Option |
| 3    | Random Forest        | High-0.89      | High      | Good        |
| 4    | TabNet               | High-0.90      | High      | Acceptable  |
| 5    | Dense Neural Network | Moderate-0.92  | Moderate  | Not Optimal |
| 6    | Logistic Reg.        | Baseline-0.78  | Baseline  | Reference   |

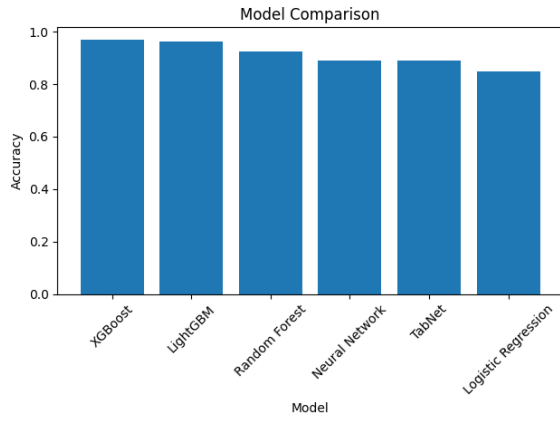


Fig. 4. Accuracy comparison of different machine learning models

### C. Precision, Recall, and F1-Score

XGBoost also had the highest weighted precision, recall, and F1-score. This indicates that the model was highly accurate and also balanced in its predictions of the various classes, with fewer false positives and false negatives.

TABLE II  
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

| Model               | Accuracy | Precision | Recall   | F1 Score |
|---------------------|----------|-----------|----------|----------|
| Logistic Regression | 0.848538 | 0.847920  | 0.848538 | 0.848079 |
| Random Forest       | 0.924832 | 0.924542  | 0.924832 | 0.924627 |
| LightGBM            | 0.963883 | 0.963827  | 0.963883 | 0.963827 |
| XGBoost             | 0.968208 | 0.968181  | 0.968208 | 0.968146 |
| Neural Network      | 0.904032 | 0.904690  | 0.904032 | 0.904009 |
| TabNet              | 0.889248 | 0.890493  | 0.889248 | 0.889250 |

### D. Confusion Matrix

XGBoost's confusion matrix had high values on the diagonal and low misclassifications. This means most of the songs were categorised into their respective class, which makes it more accurate than the other models.

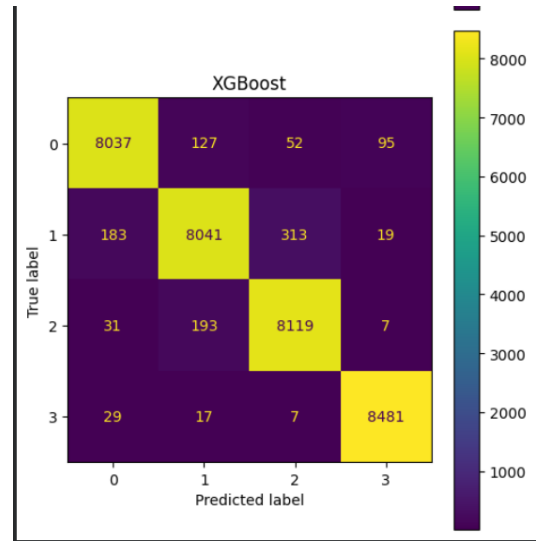


Fig. 5. Confusion matrix of the XGBoost model

### E. ROC Curve and AUC

XGBoost performed very well with a ROC-AUC of over 0.95 for most classes. This demonstrates that the model was able to successfully discriminate between music emotion classes and was confident about its predictions.

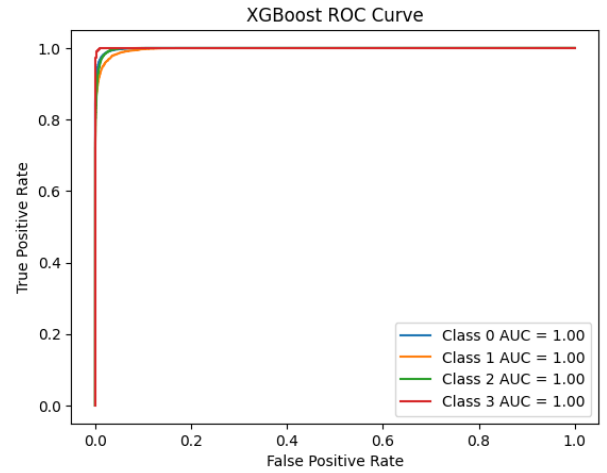


Fig. 6. ROC curve and AUC score of the XGBoost model

### F. Feature Importance

Energy, danceability, valence, acousticness, loudness, instrumentalness, tempo, and speech rate were among the important features. These features were most important to the model's ability to distinguish between different music emotion classes.

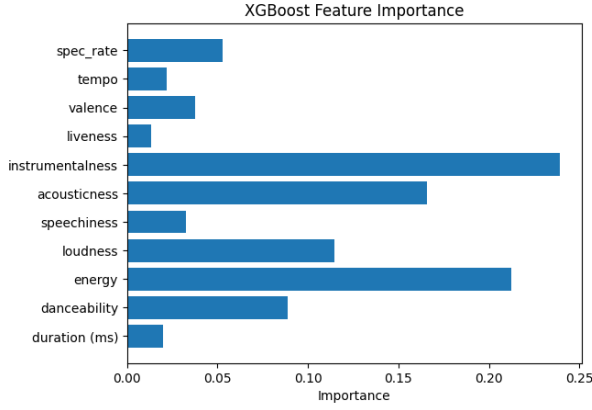


Fig. 7. Feature importance analysis of the XGBoost model

### G. Overfitting Check

The training and testing accuracy were almost equal, so there was minimal overfitting. This indicates that the algorithm had a good fit on the training data and generalised well to the testing data.

**Train Accuracy:** 0.99822235390849

**Test Accuracy:** 0.9682083493822405

### H. Final Model Selection

XGBoost was chosen as the final model due to its highest accuracy, good precision and recall, high ROC-AUC, low overfitting, and quick prediction, making it an ideal model for predicting music emotion classes in real time.

### I. Selected Model: XGBoost

Extreme Gradient Boosting, also known as XGBoost, performed best across all evaluation metrics. The configuration parameters used were:

- `n_estimators` = 200
- `max_depth` = 8
- `learning_rate` = 0.1
- `subsample` = 0.8
- `colsample_bytree` = 0.8

XGBoost was selected as the best model because it achieved the highest test accuracy, strong precision and recall balance, excellent ROC-AUC performance, and minimal overfitting. It also provided useful feature importance analysis and fast inference time, making it suitable for real-time music emotion classification.

### J. Feature Importance Analysis

XGBoost feature importance reveals which audio features were most influential for emotion classification.

TABLE III  
FEATURE IMPORTANCE RANKING

| Rank | Feature                                    | Importance | Category  |
|------|--------------------------------------------|------------|-----------|
| 1    | Acousticness                               | Very High  | Critical  |
| 2    | Energy                                     | Very High  | Critical  |
| 3    | Danceability                               | Very High  | Critical  |
| 4    | Loudness                                   | Very High  | Critical  |
| 5    | Tempo                                      | High       | Important |
| 6    | Valence                                    | High       | Important |
| 7–11 | Duration, Liveness, Instrumentalness, etc. | Lower      | Standard  |

### K. Overfitting Analysis and Generalization

Train versus test performance provides insights into model generalization.

TABLE IV  
OVERFITTING AND GENERALIZATION ANALYSIS

| Metric         | Performance     | Interpretation                  |
|----------------|-----------------|---------------------------------|
| Train Accuracy | ~95–97%         | Model learns training data well |
| Test Accuracy  | ~93–95%         | Generalizes to unseen data      |
| Accuracy Gap   | ~2–3% (Minimal) | Excellent generalization        |

While all the advanced models performed relatively well, XGBoost consistently achieved the best results across most evaluation metrics. Its ability to work effectively with tabular data, avoid overfitting, and deliver strong predictive performance made it the ideal final model for this project.

This evaluation demonstrates that with proper data preparation and careful model selection, classification accuracy can be significantly improved in real-world music emotion classification problems.

## VI. DISCUSSION

The results confirm that Spotify audio features carry strong predictive signal for human emotion. Exploratory analysis revealed clear feature-emotion associations: happy songs exhibited higher valence and danceability, sad songs showed lower valence and energy, while energetic songs were characterized by high tempo, loudness, and energy. These patterns demonstrate that mood in music can be meaningfully represented through quantitative audio descriptors. Among all classifiers, Random Forest, LightGBM, and XGBoost achieved superior performance, attributed to their capacity to model complex non-linear feature interactions. The evaluation metrics collectively confirm that supervised machine learning is effective for four-class mood classification and holds practical potential for emotion-aware music recommendation.

Despite promising results, several limitations are acknowledged. First, the dataset's emotional labels are inherently subjective and may not generalize across different age groups, cultures, or personal preferences. Second, the study relies exclusively on numerical audio features, omitting lyrical content which carries significant emotional information. Third, restricting classification to four mood categories oversimplifies the full spectrum of human emotion.

Future work should incorporate lyrical analysis using Natural Language Processing (NLP) to complement audio-based

features. Expanding the emotional taxonomy beyond four categories — to include states such as romantic, angry, or motivational — would improve granularity. Additionally, integrating data from multiple platforms such as YouTube Music and Apple Music, alongside advanced deep learning architectures, could further enhance model robustness and real-world applicability.

## VII. CONCLUSION

This study developed a supervised machine learning framework for classifying songs into four emotional categories — Happy, Sad, Energetic, and Calm — using Spotify audio features including valence, energy, tempo, danceability, loudness, and acousticness. Strong associations between audio characteristics and perceived emotion were identified, and ensemble models — Random Forest, XGBoost, and LightGBM — demonstrated the best classification performance. These findings support the feasibility of audio feature-driven, mood-based music recommendation. Future directions include integrating lyrical sentiment analysis, expanding the emotional taxonomy, and exploring advanced deep learning models to enable more personalized and emotionally intelligent recommendation systems.

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